Final Presentation & Software Demonstration

# Campus AR Navigation App

Group:
MCS 03

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Speaker: Chong Jia Yee

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Speaker: Chong Jia Yee

# Problem Statement

First day and I'm running late!

**Peter**

# Existing 2d navigation map

MazeMap

Cannot detect starting location automatically

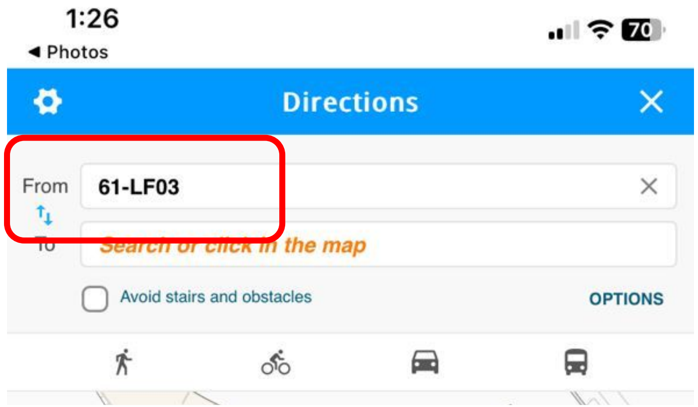
From: Choose start point

To: 6602 | Lab 6 Two

Speaker: Chong Jia Yee

# Existing 2D Navigation Map


Scan marker to indicate user's starting location

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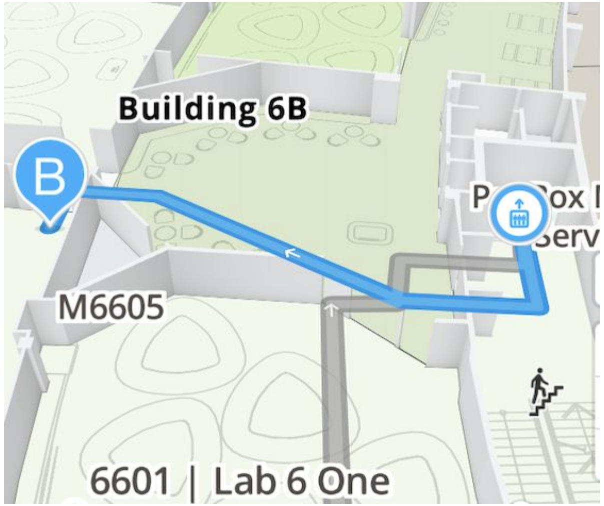
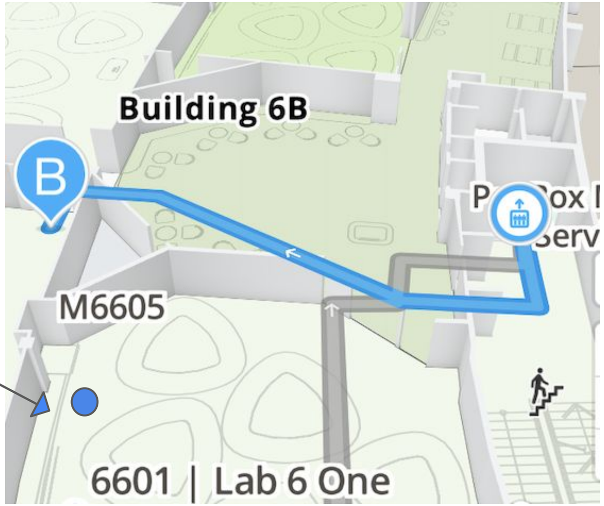
# Existing 2d navigation map

- No direction guide
- Route is fixed (will not be updated)



# Existing 2d navigation map

- What if Peter walked into a wrong path?
- He has to find the nearest marker again?



# Existing 2d navigation map

**Peter is still lost**

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Campus AR QUEST

Speaker: Chong Jia Yee

# Project Objectives

- **Real time indoor positioning** - Use beacon to accurately localise a user's indoor position (no markers scanning)
- **Active route -** Reroute when according to user's location
- **Fastest route** - use algorithm to calculate fastest route for user

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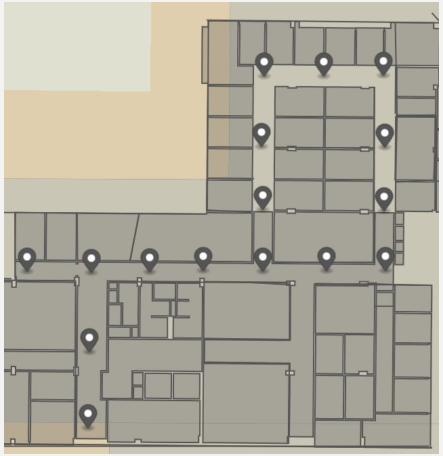
# Project Overview

Speaker: Beh Hanyu

# Final Scope

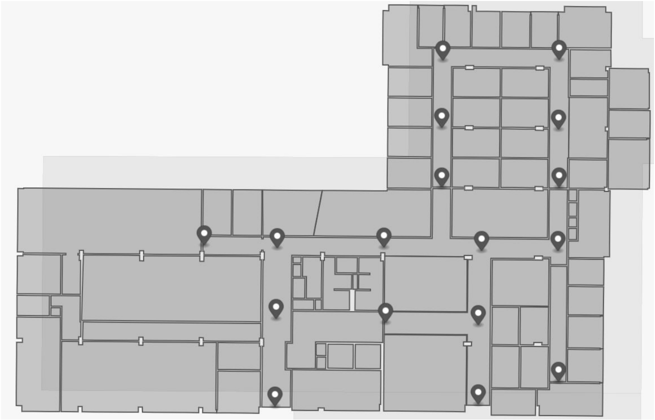


**2 floors**
Building 2, Level 3
Building 2, Level 4

Speaker: Beh Hanyu

# Limitations

Level 3 map



Level 4 map

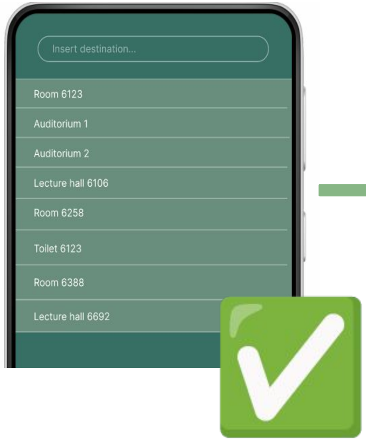
- Limited Coverage: Only implemented for Level 3 and Level 4
- Limited Number of Beacons
- Reliance on Bluetooth Beacons

Speaker: Beh Hanyu

# Project Outcome

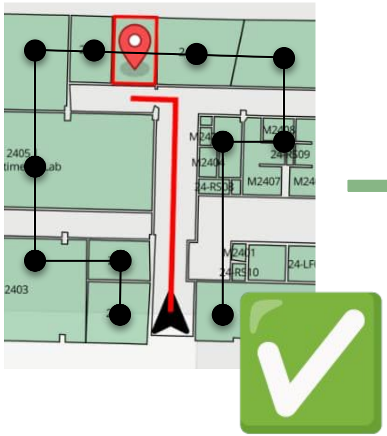
Speaker: Beh Hanyu

# What was achieved?




**Frontend**


**Localization**




**Routing**


**Voice Navigation**

Speaker: Beh Hanyu

# Methodology

Speaker: Beh Hanyu

# Software specification

android

Platform

Java

Frontend Language

IDE

python

Backend Language

SQLite

Database

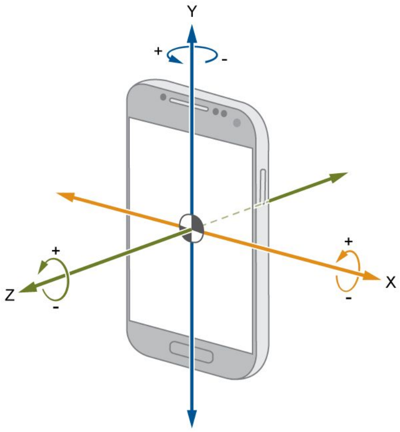
MazeMap

mapbox

Mapping SDK

# Hardware specification


Bluetooth beacons




Gyroscope

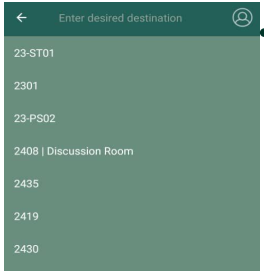

Android phone

# Deliverables

Speaker: Beh Hanyu

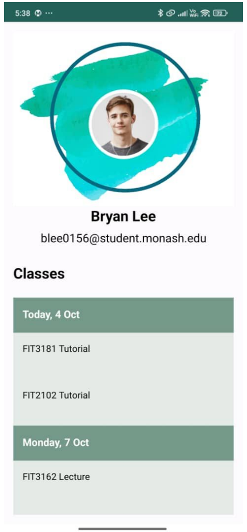
1. Automatic Indoor Position Detection & Orientation


2. Point of Interest





3. Student Account Integration

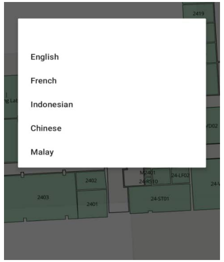


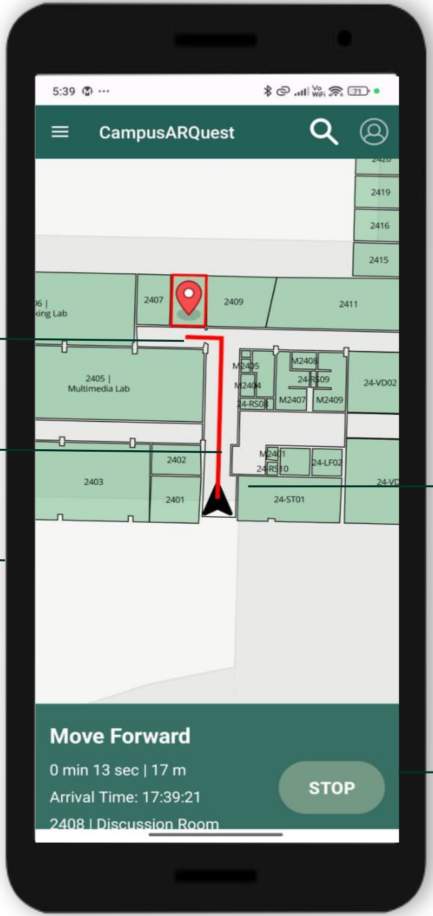

Speaker: Beh Hanyu

4. Shortest Route

5. Multi-floor Navigation

6. Voice Navigation




7. Rerouting

8. ETA and Distance Visualization

Speaker: Beh Hanyu

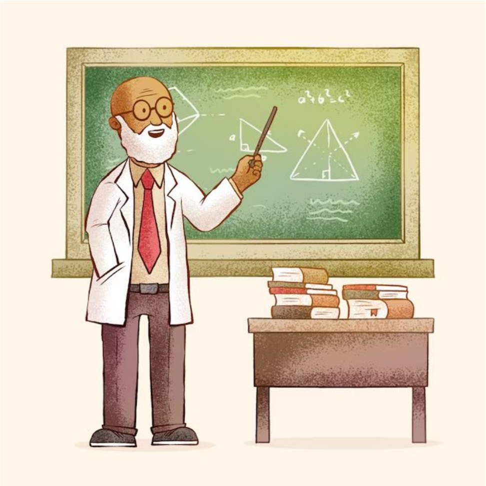
# Deliverable Benefits

Speaker: Wong Jia Cheng

# Who?

Students

Lecturers

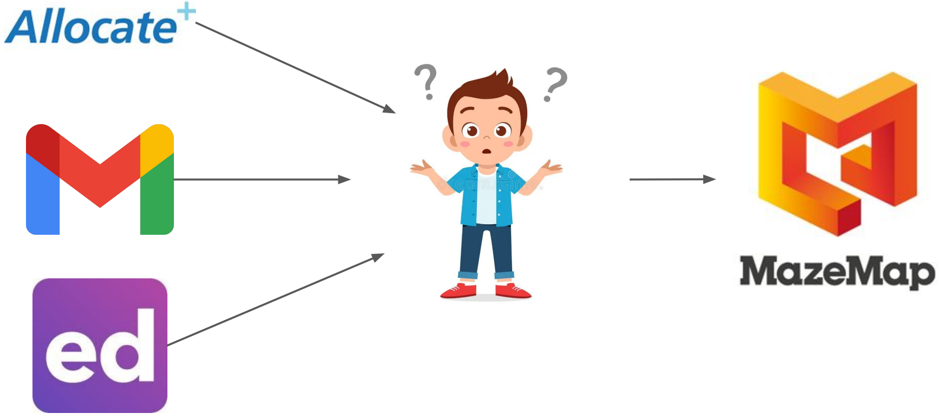


Visitors

Speaker: Wong Jia Cheng

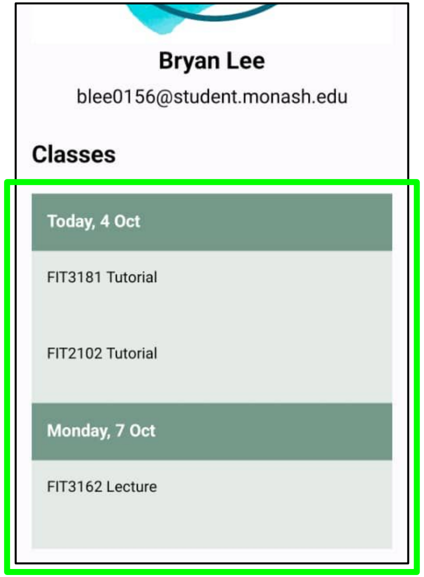
# Centralized Interface

Problem:




Time Consuming

Solution:

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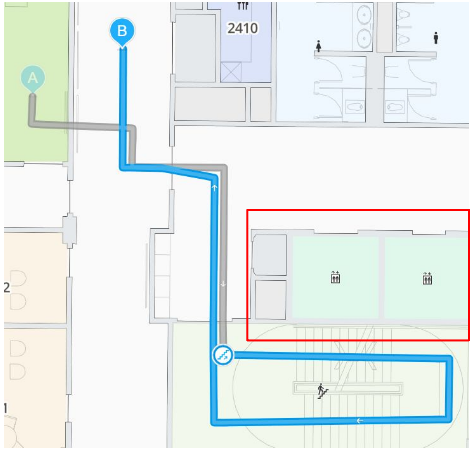
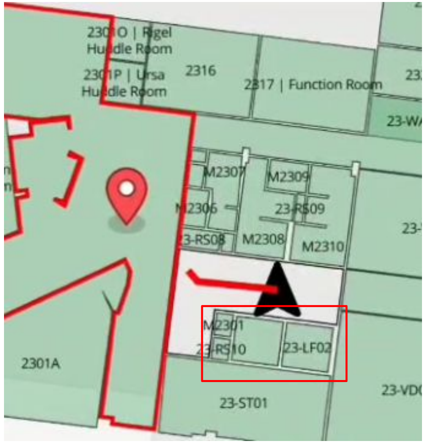
# Shortest Route

## Problem:

- Pre-defined routes.
- Inaccurate judgement of shortest route.
- Not scalable.




## Solution: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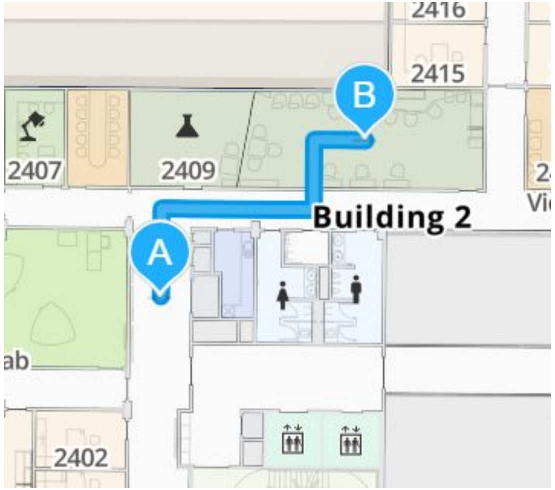
# Orientation

Problem:

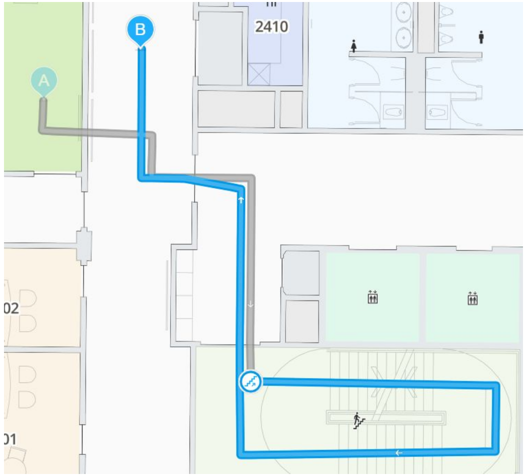



Solution:


Speaker: Wong Jia Cheng

# Multi-floor Navigation & Re-Routing

Problem:

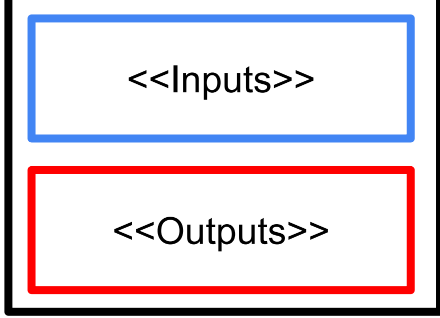


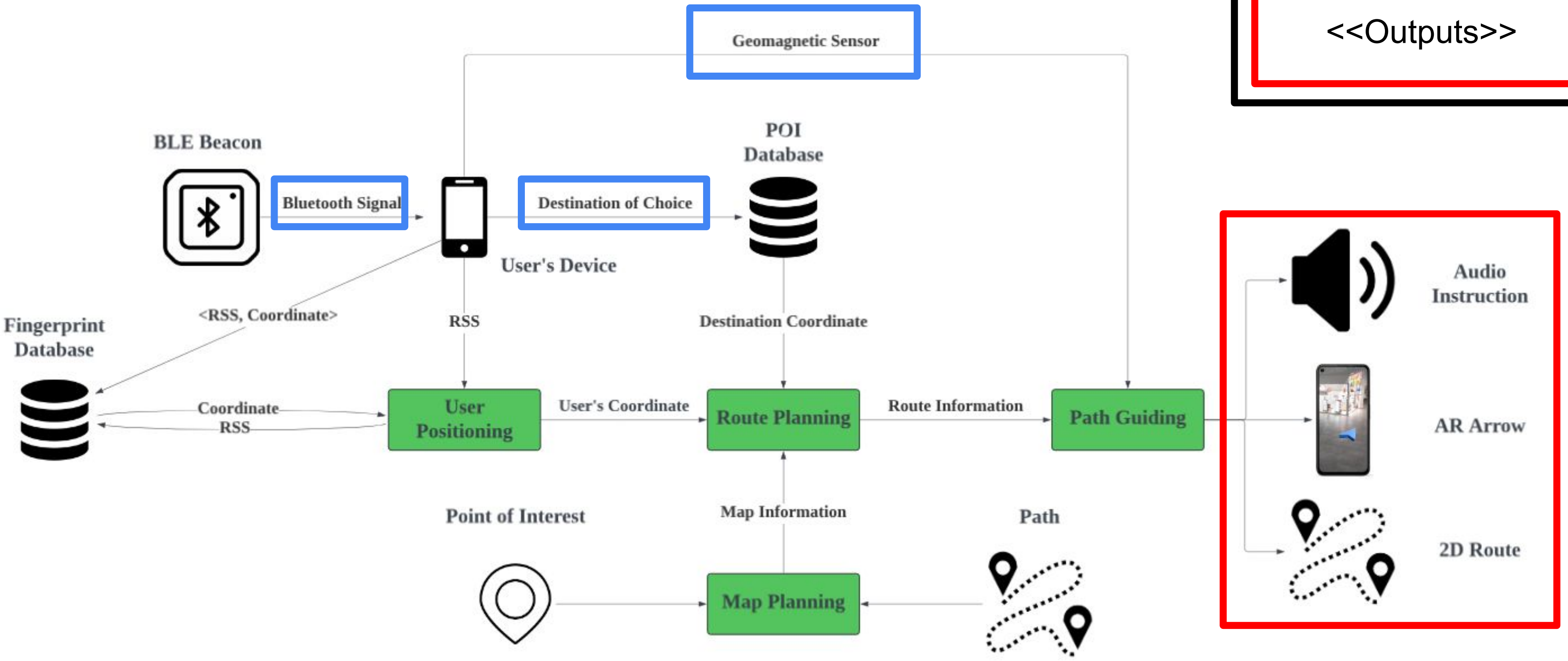

Solution:



Speaker: Wong Jia Cheng

# Project Architecture




Speaker: Wong Jia Cheng

# Demonstration

Speaker: Chong Jia Yee

# Video Demonstration

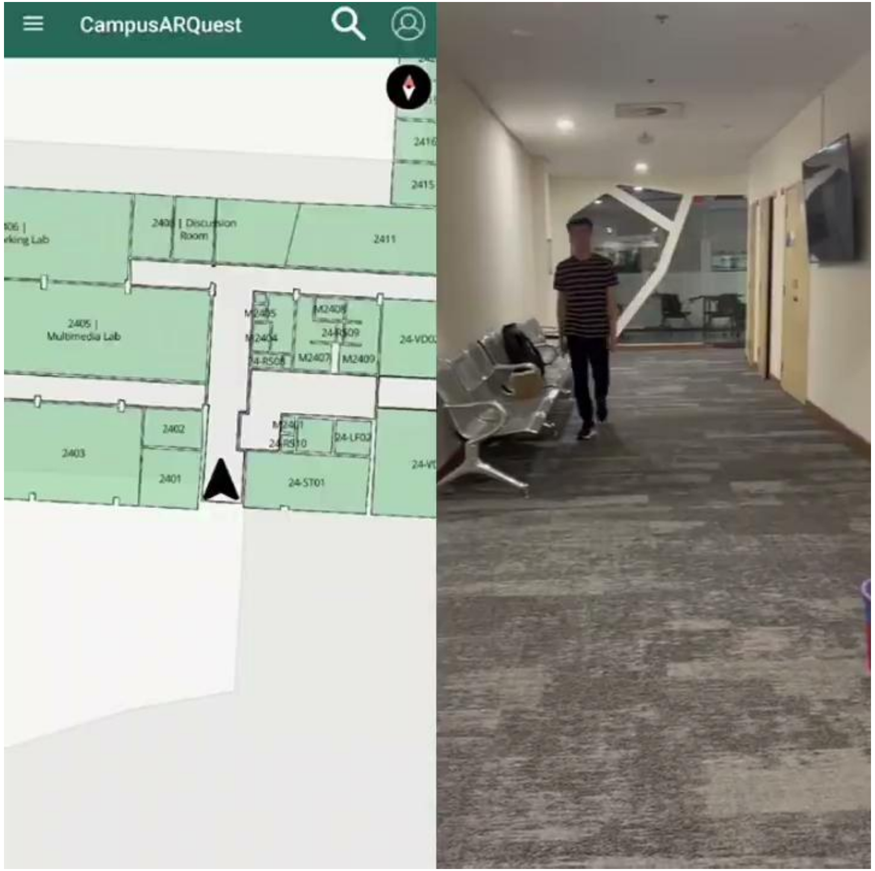
Speaker: Chong Jia Yee

# Recording Method

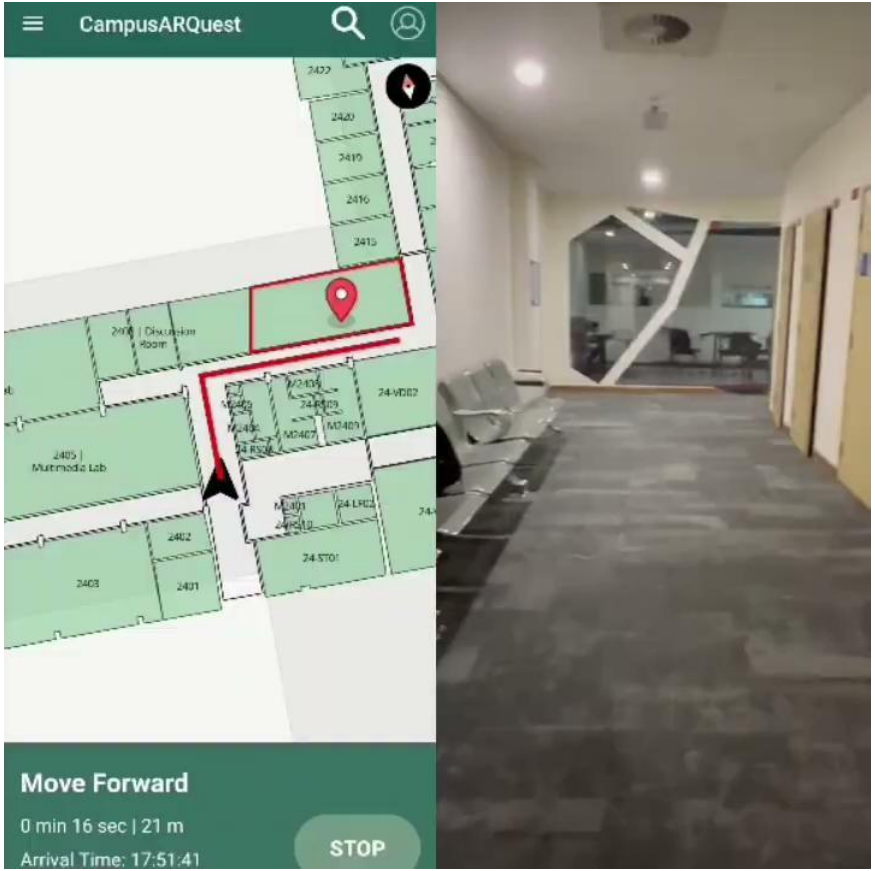
Screen record of navigation app

User's point of view

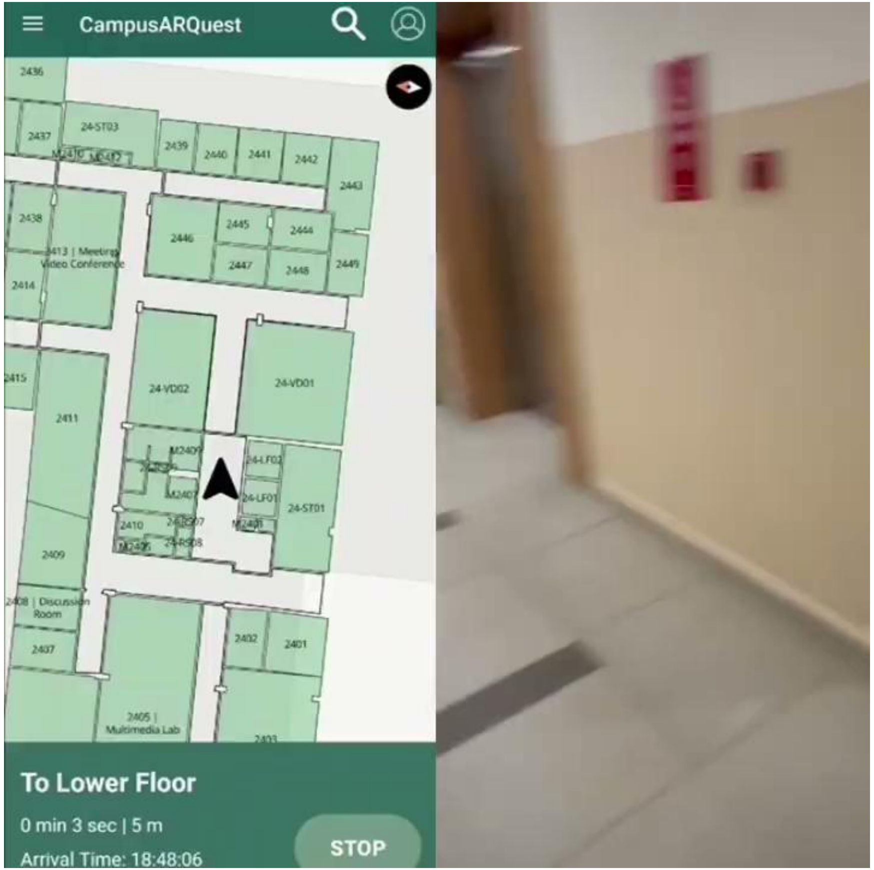
# Working Demonstration



# Automatic Rerouting



# Different Floor navigation



# Live Demonstration

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# Conclusion

Speaker: Wong Jia Cheng

## Success

- Full-fledged Mobile Application.
- Encompass general features expected from indoor navigation systems.
- Solved limitations from existing indoor navigation systems.

| RTM ID | Description | (FR / NFR) | Category | Test Case ID |
|---|---|---|---|---|
| FR1 | Application will be functioning for any indoor area of the 3rd and 4th floor of Building 2. | FR | Backend | B08 |
| FR2 | Application will display directions on screen to allow users to navigate indoor areas. | FR | Frontend | B11 |
| FR3 | The application will track the user's position and update the navigation direction in real time. | FR | Backend | W02, W06, B06-B09, B13-B14, U06, I02 |
| FR4 | The application will have audio cues that direct users with visual impairment. | FR | Frontend | B15-B16 |
| | The application will be developed with English as | | | |

## Failures & Limitations


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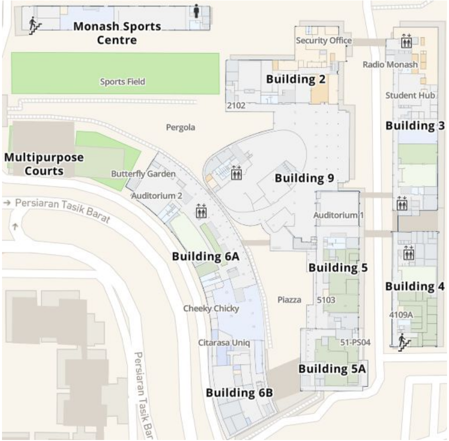
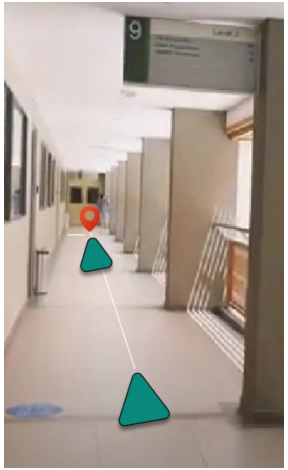
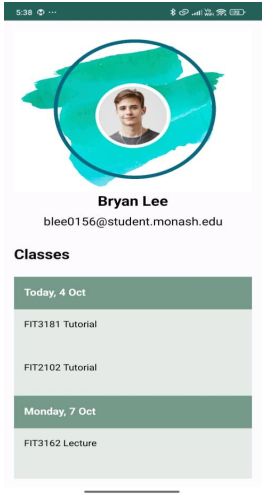
# Different Floor navigation




Extend map to cover entire campus




AR feature for camera view




More customization

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# THANK YOU